QF620 Stochastic Modelling in Finance — Session 5 Reference Cheatsheet

Stochastic Differential Equations: SDEs and martingales via Itô, the Bachelier (arithmetic) model and its call formula, the Black–Scholes (geometric) model and its call formula, the Vasicek mean-reverting model

Built from Session 5 slides (Tee Chyng Wen, SMU) · October 2026 · revised edition with intuition and full slide derivations

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Source: "Session 5: Stochastic Differential Equations" (Tee Chyng Wen, SMU, 16 slides).

How to read this sheet. Each concept entry has up to five parts: **The idea in words** (what the concept is and why it is there), the formula with a shaded where box naming every symbol, **Derivation** (the slide's argument line by line, with the reason for each step in brown), **Worked example**, and **Exam angle** (traps and what to be able to reproduce). Slide numbers are PDF page numbers, shown as [S#]. Material not on the slides is tagged *[Beyond slides]*. △**!** marks a slide ambiguity or error, with the corrected version.

What you must be able to do after this session: use Itô's formula to decide whether a process is a martingale (zero drift); solve the arithmetic, geometric and Vasicek SDEs; derive the Bachelier and Black–

Scholes call prices by splitting the integral at x^* and completing the square; get the mean, variance and distribution of the Vasicek short rate via the Itô isometry.

Notation used throughout (slide conventions)

Symbol	Meaning	Notes
$W_t,\ \tilde{W}_t$	(independent) Brownian motions	[S5]
S_t, S_0, S_T	stock price now / at 0 / at maturity	
$K,\ T,\ r$	strike; maturity; continuously compounded risk-free rate	discount e^{-rT}
σ	volatility: price units (Bachelier), percentage (Black–Scholes)	△! same letter [S10]
V_c	European call price	[S7]
$\Phi,\ \phi$	standard normal CDF, PDF	$\phi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$
x^*	exercise threshold for the standard normal variable	[S7, S12]
d_1, d_2	Black–Scholes arguments	[S14]
$r_t,\ \kappa,\ \theta$	short rate; mean-reversion speed; long-run mean	Vasicek [S15]
log	natural logarithm	slides write log for ln

Notation trap: on [S15–S16] r_t is a random short rate, while in the option formulas r is a constant discount rate. σ in Bachelier is in price units ($\approx \sigma_{BS} \times S_0$).

0. Topic map

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PREREQUISITES: S1 normal MGF, completing the square ; S3 BM, martingales ; S4 Itô's formula, isometry
    ▼
SESSION 5├─
 HISTORY & MARTINGALE PROCESS [–S2S5]│
     ├─ Bru & Yor on Doeblin ; Itô calculus took 25 years to be accepted        [–S2S4]│
     └─ six processes: Itô SDE ; martingale iff zero drift                      [S5]├─
 ARITHMETIC: BACHELIER (1900) [–S6S8]│
     ├─ S_T = ₀S + σW_T ~ N(₀S, σ²T) ; negative prices possible                [S6]│
     ├─ V_c = e–^{rT}E[(S_T – K)⁺] ; exercise when x > x* = (K – ₀S)σ√/(T)       [S7]│
     └─ V_c = e–^{rT}[(₀–SK)Φ(d) + σ√T φ(d)] ; ATM σ√(Tπ/2)                      [S8]├─
 GEOMETRIC: –BLACKSCHOLES (1973) [–S9S14]│
     ├─ S_T = ₀Sexp[(r – σ½²)T + σW_T] ; paths vs Bachelier                     [–S9S10]│
     ├─ dS = rS dt + σS dW solved via Itô on log S                              [S11]│
     └─ x* ; completing the square ; V_c = ₀ΦS(₁d) – Ke–^{rTΦ}(₂d)              [–S12S14]└─
 MEAN REVERSION: VASICEK (1977) [–S15S16]
      ├─ dr = κθ( – r)dt + σdW ; Itô on eκ^{t}r_t ; explicit solution            [S15]
      └─ r_t ~ N(₀re–κ^{t} + θ(1 – e–κ^{t}), σ²(1 – e–κ^{2t})κ/(2))              [S16]
    ▼
LATER: S6 implied volatility, smile, displaced diffusion, stochastic volatility
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The storyline of the session.

- **Itô turns "is it a martingale?" into "is the** $\mathrm{d}t$ **coefficient zero?"**. Six examples drill this.
- **Solving an SDE** means finding a transformation that makes the right-hand side free of the unknown: none needed for arithmetic BM, $\log S$ for geometric BM, the integrating factor $e^{\kappa t} r_t$ for Vasicek. Then integrate.
- **Pricing** once S_T is a function of one standard normal x: any European price is a one-dimensional integral $e^{-rT} \int \text{payoff}(x)\phi(x)\,\mathrm{d}x$. For a call, cut the integral at the exercise boundary x^* and evaluate the two pieces (by $\int x\phi = \phi$ for Bachelier, by completing the square for Black–Scholes).

1 Concept entries

1.1 History of stochastic calculus [S2–S4]

- Bru and Yor (2002), on Wolfgang Doeblin: stochastic processes took a tortuous path through the 20th century; pioneers (Bachelier, Lévy, Itô) were original and took long to be accepted [S2].

- Brownian motion belongs to many classes at once: Gaussian, independent increments, Markov, martingale, semimartingale [S3].
- Itô's calculus took 25 years (1944–1969) to be accepted; 1969 is McKean's *Stochastic Integrals* [S4].

1.2 SDEs and martingales [S5]

The idea in words

Apply Itô to write $\mathrm{d}X_t = a_t\,\mathrm{d}t + b_t\,\mathrm{d}W_t$. The $\mathrm{d}W$ part integrates to a stochastic integral with mean zero (Session 4), so it adds no expected change. The $\mathrm{d}t$ part is a predictable drift. Hence X is a martingale exactly when the drift a_t is zero. This is much faster than the "add and subtract W_s" method of Session 3, and it is how you check a candidate process in every later session (e.g. "is the discounted price a $\mathbb{Q}$-martingale?").

Rule *[Beyond slides]*: for $X_t = g(t, W_t)$, Itô with $\mu = 0$, $\sigma = 1$ gives

$$\mathrm{d}g(t, W_t) = \Big(g_t + \tfrac{1}{2}g_{xx}\Big)\,\mathrm{d}t + g_x\,\mathrm{d}W_t; \qquad \text{martingale} \iff g_t + \tfrac{1}{2}g_{xx} = 0.$$

For two independent BMs add $\frac{1}{2}g_{yy}\,\mathrm{d}t$ and $g_y\,\mathrm{d}\tilde{W}$ ($\mathrm{d}W\,\mathrm{d}\tilde{W} = 0$).

The six slide examples, working filled in (slide gives only yes/no)

(1) $X = W_t^2$:	$g_x = 2W,\ g_{xx} = 2 \Rightarrow \mathrm{d}X = \mathrm{d}t + 2W_t\,\mathrm{d}W_t$	(drift 1: no)
(2) $X = 2 + t + e^{W_t}$:	$g_t = 1,\ g_x = g_{xx} = e^W \Rightarrow \mathrm{d}X = \big(1 + \frac{1}{2}e^{W_t}\big)\,\mathrm{d}t + e^{W_t}\,\mathrm{d}W_t$	(no)
(3) $X = W_t^2 + \tilde{W}_t^2$:	$\mathrm{d}X = (1+1)\,\mathrm{d}t + 2W_t\,\mathrm{d}W_t + 2\tilde{W}_t\,\mathrm{d}\tilde{W}_t$	(drift 2: no)
(4) $X = W_t^2 - t$:	$\mathrm{d}X = \mathrm{d}t + 2W_t\,\mathrm{d}W_t - \mathrm{d}t = 2W_t\,\mathrm{d}W_t$	(yes)
(5) $X = W_t^3$:	$g_x = 3W^2,\ g_{xx} = 6W \Rightarrow \mathrm{d}X = 3W_t\,\mathrm{d}t + 3W_t^2\,\mathrm{d}W_t$	(drift $3W_t$: no)
(6) $X = e^{\theta W_t - \frac{1}{2}\theta^2 t}$:	$g_t = -\frac{1}{2}\theta^2 X,\ g_x = \theta X,\ g_{xx} = \theta^2 X$	
	$\mathrm{d}X = \big(-\frac{1}{2}\theta^2 + \frac{1}{2}\theta^2\big)X_t\,\mathrm{d}t + \theta X_t\,\mathrm{d}W_t = \theta X_t\,\mathrm{d}W_t$	(yes)

In (6) the explicit $-\frac{1}{2}\theta^2 t$ produces a drift $-\frac{1}{2}\theta^2 X$ that exactly cancels the Itô term $+\frac{1}{2}\theta^2 X$: the exponential martingale of Session 3.

Exam angle

△! *[Beyond slides]* Zero drift strictly gives a *local* martingale; it is a true martingale when $\mathbb{E}\int_0^T b_t^2\,\mathrm{d}t < \infty$, true for all slide examples. Compare (5) with Session 3: drift $3W_t$ is why $\mathbb{E}_s[W_t^3] = W_s^3 + 3(t-s)W_s$.

1.3 Bachelier model: arithmetic Brownian motion [S6–S7]

The idea in words

Bachelier (1900) modelled the price as a symmetric random walk: additive normal shocks of the same size whatever the price level. Solving is trivial (integrate both sides). Pricing: write S_T as a function of one standard normal x, write the call price as an integral against $\phi(x)$, and note the payoff is zero below a threshold x^*: that is where the call starts to finish in the money.

$$\mathrm{d}S_t = \sigma\,\mathrm{d}W_t \implies S_T = S_0 + \sigma W_T, \quad W_T \sim \mathcal{N}(0, T), \quad S_T \sim \mathcal{N}(S_0, \sigma^2 T).$$

where

σ	absolute (normal) volatility, in price units per $\sqrt{\text{year}}$.
$\sigma^2 T$	variance grows linearly in time.

Shortcoming [S6]: no lower bound at 0, so negative stock prices have positive probability; reasonable for interest rates (which can be negative).

Setting up the call integral [S7]

$$V_c = e^{-rT}\mathbb{E}\big[(S_T-K)^+\big] = e^{-rT}\mathbb{E}\big[(S_0+\sigma W_T-K)^+\big] \quad \text{(risk-neutral price)}$$
$$= \frac{e^{-rT}}{\sqrt{2\pi}}\int_{-\infty}^{\infty}\big(S_0+\sigma\sqrt{T}x-K\big)^+ e^{-\frac{x^2}{2}}\,\mathrm{d}x \quad (W_T=\sqrt{T}x,\ x\sim\mathcal{N}(0,1))$$
$$S_0+\sigma\sqrt{T}x-K>0 \iff x>\frac{K-S_0}{\sigma\sqrt{T}}=x^* \quad \text{(in-the-money region)}$$

where

e^{-rT}	discount factor from T back to today.
x^*	critical value of x above which the call finishes in the money; $\sigma\sqrt{T}$ standardises the distance $K-S_0$.

Exam angle

△! [S6–S7] The model has zero drift but discounts at r: only risk-neutral if $r=0$, or if S is read as the forward price. Under $\mathbb{Q}$ with a spot stock the drift should be rS_t *[Beyond slides]*.

1.4 Bachelier call formula [S8]

The idea in words

Above x^* the payoff is linear in x: a constant part (S_0-K) and a part proportional to x. The constant part integrates against ϕ to a probability, $\Phi(-x^*)$. The linear part integrates exactly because $x\phi(x)$ has antiderivative $-\phi(x)$. So the price is "intrinsic value × probability of exercise + volatility × density". At the money the first term vanishes and the price is just $\sigma\sqrt{T/2\pi}\approx 0.4\sigma\sqrt{T}$.

Evaluating the integral [S8] — every line of the slide

$$V_c = \frac{e^{-rT}}{\sqrt{2\pi}}\int_{x^*}^{\infty}\big(S_0+\sigma\sqrt{T}x-K\big)e^{-\frac{x^2}{2}}\,\mathrm{d}x \quad \text{(drop } (\cdot)^+ \text{ on } x>x^*)$$
$$= \frac{e^{-rT}}{\sqrt{2\pi}}\int_{x^*}^{\infty}(S_0-K)e^{-\frac{x^2}{2}}\,\mathrm{d}x + \frac{e^{-rT}}{\sqrt{2\pi}}\int_{x^*}^{\infty}\sigma\sqrt{T}x\,e^{-\frac{x^2}{2}}\,\mathrm{d}x \quad \text{(split)}$$
$$= e^{-rT}(S_0-K)\big[\Phi(\infty)-\Phi(x^*)\big] - \frac{e^{-rT}\sigma\sqrt{T}}{\sqrt{2\pi}}\Big[e^{-\frac{x^2}{2}}\Big]_{x^*}^{\infty} \quad \big(\tfrac{\mathrm{d}}{\mathrm{d}x}e^{-x^2/2}=-xe^{-x^2/2}\big)$$
$$= e^{-rT}\Big[(S_0-K)\Phi(-x^*)+\frac{\sigma\sqrt{T}}{\sqrt{2\pi}}e^{-\frac{(x^*)^2}{2}}\Big] \quad \big(1-\Phi(x^*)=\Phi(-x^*)\big)$$
$$= e^{-rT}\Big[(S_0-K)\Phi(-x^*)+\frac{\sigma\sqrt{T}}{\sqrt{2\pi}}e^{-\frac{(-x^*)^2}{2}}\Big] \quad \text{(even function)}$$
$$= e^{-rT}\Big[(S_0-K)\,\Phi\Big(\frac{S_0-K}{\sigma\sqrt{T}}\Big)+\sigma\sqrt{T}\,\phi\Big(\frac{S_0-K}{\sigma\sqrt{T}}\Big)\Big] \quad \big(-x^*=\tfrac{S_0-K}{\sigma\sqrt{T}}\big)$$

ATM ($K=S_0$): Φ-term vanishes, $\phi(0)=\frac{1}{\sqrt{2\pi}}$, so $V_c=e^{-rT}\sigma\sqrt{\frac{T}{2\pi}}$.

where

$\frac{S_0-K}{\sigma\sqrt{T}}$	moneyness in standard deviations ($= -x^*$).
$(S_0-K)\Phi(\cdot)$	intrinsic value weighted by the probability of exercise.
$\sigma\sqrt{T}\phi(\cdot)$	time value from the spread of outcomes; the whole price at the money.

[Beyond slides] Put by parity: $V_p = V_c - e^{-rT}(S_0-K) = e^{-rT}[(K-S_0)\Phi(-d) + \sigma\sqrt{T}\phi(d)]$, $d = \frac{S_0-K}{\sigma\sqrt{T}}$.

1.5 Black–Scholes model: geometric Brownian motion [S9–S10]

The idea in words

Black and Scholes (1973) made shocks *proportional* to the price: percentage returns are normal, so the price is lognormal and always positive. The drift of the log is $r - \frac{1}{2}\sigma^2$ so that the price itself grows at r on average (convexity correction, Sessions 1–2). Compare the paths: when the price is high, Black–Scholes moves are big; when low, they shrink, so the price never crosses zero. Bachelier moves are the same size everywhere.

$$S_T = S_0 \exp\left[\left(r - \frac{\sigma^2}{2}\right)T + \sigma W_T\right], \qquad \log\frac{S_T}{S_0} \sim \mathcal{N}\left(\left(r-\frac{\sigma^2}{2}\right)T,\ \sigma^2 T\right), \qquad \mathbb{E}[S_T] = S_0 e^{rT}.$$

Verifying $\mathbb{E}[S_T] = S_0 e^{rT}$ [S9]

$\mathbb{E}[S_T] = S_0 e^{(r-\frac{1}{2}\sigma^2)T}\mathbb{E}[e^{\sigma W_T}] = S_0 e^{(r-\frac{1}{2}\sigma^2)T} e^{\frac{1}{2}\sigma^2 T} = S_0 e^{rT}$ (normal MGF with variance T). Alternative form: $\log S_T = \log S_0 + (r - \frac{\sigma^2}{2})T + \sigma\sqrt{T}\,\mathcal{N}(0,1)$.

Sample paths [S10] (same Brownian path):

$$\text{BS: } S_{t+\Delta t} = S_t e^{(r-\frac{\sigma^2}{2})\Delta t + \sigma(W_{t+\Delta t} - W_t)}, \qquad \text{Bachelier: } S_{t+\Delta t} = S_t + \sigma(W_{t+\Delta t} - W_t).$$

Slide question [S10] (answered here): with matched volatility near S_0 the paths track closely. When S rises, BS moves get bigger; when S falls they shrink and S stays positive, whereas Bachelier keeps the same absolute size and can go negative. BS outcomes are lognormal and right-skewed; Bachelier outcomes are symmetric.

Exam angle

△! [S10] uses the same σ in both recursions; to be comparable the Bachelier σ must be in price units, roughly $\sigma_{BS} S_0$.

1.6 Solving the Black–Scholes SDE [S11]

The idea in words

$\mathrm{d}S = rS\,\mathrm{d}t + \sigma S\,\mathrm{d}W$ cannot be integrated directly because S appears on both sides. Dividing by S suggests the log: $\mathrm{d}S/S$ looks like $\mathrm{d}\log S$. By Itô, $\mathrm{d}\log S$ has *constant* coefficients (plus the $-\frac{1}{2}\sigma^2$ Itô correction), so it is an arithmetic BM, which integrates trivially. Exponentiate to get S_T.

Itô on $X_t = \log S_t$ [S11]

$f(S) = \log S$, $f' = \frac{1}{S}$, $f'' = -\frac{1}{S^2}$; drift of S is $\mu_t = rS_t$, volatility $\sigma_t = \sigma S_t$.

$$\begin{aligned}
\mathrm{d}X_t &= \left(rS_t \cdot \frac{1}{S_t} + \frac{1}{2}\sigma^2 S_t^2 \cdot \left(-\frac{1}{S_t^2}\right)\right)\mathrm{d}t + \sigma S_t \cdot \frac{1}{S_t}\,\mathrm{d}W_t && (\mu f' + \tfrac{1}{2}\sigma^2 f'';\ \sigma f') \\
&= \left(r - \frac{\sigma^2}{2}\right)\mathrm{d}t + \sigma\,\mathrm{d}W_t && \text{(constant coefficients)} \\
X_T - X_0 &= \left(r - \frac{\sigma^2}{2}\right)T + \sigma W_T && \text{(integrate } 0 \to T;\ W_0 = 0) \\
S_T &= S_0 \exp\left[\left(r - \frac{\sigma^2}{2}\right)T + \sigma W_T\right] && (X = \log S, \text{ exponentiate})
\end{aligned}$$

1.7 Black–Scholes call: setting up [S12]

The integral and the exercise region [S12]

With $W_T = \sqrt{T}x$, $x \sim \mathcal{N}(0,1)$:

$$V_c = e^{-rT}\mathbb{E}[(S_T - K)^+] = \frac{e^{-rT}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \left(S_0 e^{(r-\frac{\sigma^2}{2})T+\sigma\sqrt{T}x} - K\right)^+ e^{-\frac{x^2}{2}}\,\mathrm{d}x$$

$$S_T > K \iff (r-\tfrac{\sigma^2}{2})T + \sigma\sqrt{T}x > \log\tfrac{K}{S_0} \qquad \text{(take logs)}$$

$$\iff x > \frac{\log\frac{K}{S_0} - (r-\frac{\sigma^2}{2})T}{\sigma\sqrt{T}} = x^* \qquad \text{(solve for } x\text{)}$$

$$V_c = \frac{e^{-rT}}{\sqrt{2\pi}} \int_{x^*}^{\infty} \left(S_0 e^{(r-\frac{\sigma^2}{2})T+\sigma\sqrt{T}x} - K\right) e^{-\frac{x^2}{2}}\,\mathrm{d}x \qquad \text{(drop } (\cdot)^+\text{)}$$

where

$\log(K/S_0)$	log-distance from spot up to the strike.
$(r-\frac{1}{2}\sigma^2)T$	expected log drift, which works in favour of exercise.
x^*	so $\mathbb{Q}(\text{exercise}) = \Phi(-x^*) = \Phi(d_2)$.

1.8 Black–Scholes call: completing the square [S13–S14]

The idea in words

The strike term is easy: Ke^{-rT} times the probability $x > x^*$. The stock term has $e^{\sigma\sqrt{T}x}$ multiplying $e^{-x^2/2}$: an exponential of a quadratic. As in Session 1's MGF, complete the square: $-\frac{x^2}{2} + \sigma\sqrt{T}x = -\frac{(x-\sigma\sqrt{T})^2}{2} + \frac{\sigma^2 T}{2}$. The constant $e^{\sigma^2 T/2}$ cancels the $e^{-\sigma^2 T/2}$ in front and e^{rT} cancels e^{-rT}; what is left is a normal density shifted by $\sigma\sqrt{T}$, which integrates to a Φ with argument shifted by $\sigma\sqrt{T}$. That shift is exactly $d_1 = d_2 + \sigma\sqrt{T}$.

Every line of [S13]

$$V_c = \frac{e^{-rT}}{\sqrt{2\pi}} \int_{x^*}^{\infty} S_0 e^{(r-\frac{\sigma^2}{2})T+\sigma\sqrt{T}x-\frac{x^2}{2}}\,\mathrm{d}x - \frac{e^{-rT}}{\sqrt{2\pi}} \int_{x^*}^{\infty} Ke^{-\frac{x^2}{2}}\,\mathrm{d}x \qquad \text{(split)}$$

$$= \frac{S_0 e^{-\frac{\sigma^2 T}{2}}}{\sqrt{2\pi}} \int_{x^*}^{\infty} e^{-\frac{x^2-2\sigma\sqrt{T}x}{2}}\,\mathrm{d}x - \frac{Ke^{-rT}}{\sqrt{2\pi}} \int_{x^*}^{\infty} e^{-\frac{x^2}{2}}\,\mathrm{d}x \qquad (e^{-rT}e^{rT}=1)$$

$$= \frac{S_0 e^{-\frac{\sigma^2 T}{2}}}{\sqrt{2\pi}} \int_{x^*}^{\infty} e^{-\frac{x^2-2\sigma\sqrt{T}x+\sigma^2T-\sigma^2T}{2}}\,\mathrm{d}x - \cdots \qquad \text{(add and subtract } \sigma^2T\text{)}$$

$$= \frac{S_0 e^{-\frac{\sigma^2 T}{2}}}{\sqrt{2\pi}} \int_{x^*}^{\infty} e^{-\frac{(x-\sigma\sqrt{T})^2}{2}} e^{\frac{\sigma^2 T}{2}}\,\mathrm{d}x - \cdots \qquad \text{(perfect square)}$$

$$= \frac{S_0}{\sqrt{2\pi}} \int_{x^*}^{\infty} e^{-\frac{(x-\sigma\sqrt{T})^2}{2}}\,\mathrm{d}x - \frac{Ke^{-rT}}{\sqrt{2\pi}} \int_{x^*}^{\infty} e^{-\frac{x^2}{2}}\,\mathrm{d}x \qquad \text{(constants cancel)}$$

Evaluating and naming d_1, d_2 [S14]

In the first integral substitute $y = x - \sigma\sqrt{T}$ (lower limit $x^* - \sigma\sqrt{T}$):

$$V_c = S_0\left[\Phi(\infty) - \Phi(x^* - \sigma\sqrt{T})\right] - Ke^{-rT}\left[\Phi(\infty) - \Phi(x^*)\right]$$

$$= S_0\Phi(-x^* + \sigma\sqrt{T}) - Ke^{-rT}\Phi(-x^*) \qquad (1-\Phi(a) = \Phi(-a))$$

$$-x^* = \frac{\log\frac{S_0}{K} + (r-\frac{\sigma^2}{2})T}{\sigma\sqrt{T}} =: d_2, \qquad -x^* + \sigma\sqrt{T} = \frac{\log\frac{S_0}{K} + (r+\frac{\sigma^2}{2})T}{\sigma\sqrt{T}} =: d_1 \qquad (\sigma\sqrt{T} = \tfrac{\sigma^2 T}{\sigma\sqrt{T}})$$

$$\boxed{V_c = S_0\Phi(d_1) - Ke^{-rT}\Phi(d_2), \qquad d_1 = \frac{\log\frac{S_0}{K} + (r+\frac{\sigma^2}{2})T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}.}$$

where

$\Phi(d_2)$	$\mathbb{Q}(S_T > K)$: risk-neutral exercise probability.
$\Phi(d_1)$	exercise probability under the stock measure; also the call delta $\partial V_c/\partial S_0$ *[Beyond slides]*.
$S_0\Phi(d_1)$	present value of receiving the stock when exercised.
$Ke^{-rT}\Phi(d_2)$	present value of paying the strike when exercised.

[Beyond slides] Put: $V_p = Ke^{-rT}\Phi(-d_2) - S_0\Phi(-d_1)$; parity $V_c - V_p = S_0 - Ke^{-rT}$. Black (forward) form: $V_c = e^{-rT}[F\Phi(d_1) - K\Phi(d_2)]$, $F = S_0e^{rT}$, $d_1 = \frac{\log(F/K)+\frac{1}{2}\sigma^2T}{\sigma\sqrt{T}}$.

Exam angle

The derivation is a classic exam question. Checklist: (1) S_T in terms of x; (2) find x^*; (3) split; (4) complete the square in the stock term; (5) substitute $y = x - \sigma\sqrt{T}$; (6) use $1 - \Phi(a) = \Phi(-a)$; (7) identify $d_2 = -x^*$, $d_1 = d_2 + \sigma\sqrt{T}$.

1.9 Vasicek mean-reverting model [S15]

The idea in words

Interest rates do not wander off forever: they are pulled back to a long-run level θ. The drift $\kappa(\theta - r_t)$ is positive below θ and negative above, harder the further away. To solve, we need to remove r_t from the drift. Multiplying by the integrating factor $e^{\kappa t}$ does it (as for the ODE $y' = -\kappa y$): $\mathrm{d}(e^{\kappa t}r_t)$ has no r_t on the right, so it integrates directly. The solution shows exponentially fading memory of r_0, a pull towards θ, and shocks damped by their age.

$$\mathrm{d}r_t = \kappa(\theta - r_t)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t.$$

where

θ	long-run mean level.
$\kappa > 0$	speed of mean reversion.
$\theta - r_t$	positive below the mean (drift up), negative above (drift down).
σ	absolute volatility of the rate (normal model).

Solving with the integrating factor [S15]

Itô on $X_t = f(t, r_t) = e^{\kappa t}r_t$: $f_t = \kappa e^{\kappa t}r$, $f_r = e^{\kappa t}$, $f_{rr} = 0$ (no Itô term).

$$\begin{aligned}
\mathrm{d}(e^{\kappa t}r_t) &= \kappa e^{\kappa t}r_t\,\mathrm{d}t + e^{\kappa t}\,\mathrm{d}r_t && \text{(Itô with time)}\\
&= \kappa e^{\kappa t}r_t\,\mathrm{d}t + e^{\kappa t}\big[\kappa\theta - \kappa r_t\big]\,\mathrm{d}t + \sigma e^{\kappa t}\,\mathrm{d}W_t && \text{(substitute } \mathrm{d}r_t\text{)}\\
&= \kappa\theta e^{\kappa t}\,\mathrm{d}t + \sigma e^{\kappa t}\,\mathrm{d}W_t && (\pm\kappa e^{\kappa t}r_t \text{ cancel})\\
e^{\kappa t}r_t - r_0 &= \int_0^t \kappa\theta e^{\kappa u}\,\mathrm{d}u + \sigma\int_0^t e^{\kappa u}\,\mathrm{d}W_u = \theta(e^{\kappa t} - 1) + \sigma\int_0^t e^{\kappa u}\,\mathrm{d}W_u && \text{(integrate } 0 \to t\text{)}\\
r_t &= r_0e^{-\kappa t} + \theta\big(1 - e^{-\kappa t}\big) + \sigma\int_0^t e^{\kappa(u-t)}\,\mathrm{d}W_u && \text{(multiply by } e^{-\kappa t}\text{)}
\end{aligned}$$

where

$r_0e^{-\kappa t}$	memory of the starting rate, decaying exponentially.
$\theta(1 - e^{-\kappa t})$	pull toward the long-run mean.
$\sigma\int e^{\kappa(u-t)}\,\mathrm{d}W_u$	accumulated shocks; a shock at time u is damped by $e^{-\kappa(t-u)}$.

1.10 Vasicek distribution [S16]

Mean, variance, distribution [S16]

$$\mathbb{E}[r_t] = r_0 e^{-\kappa t} + \theta(1 - e^{-\kappa t}) \qquad \text{(stochastic integral has mean 0)}$$

$$V[r_t] = \mathbb{E}\Big[\Big(\sigma \int_0^t e^{\kappa(u-t)}\, \mathrm{d}W_u\Big)^2\Big] \qquad \text{(only the random part)}$$

$$= \sigma^2 \int_0^t e^{2\kappa(u-t)}\, \mathrm{d}u \qquad \text{(Itô isometry; deterministic integrand)}$$

$$= \sigma^2 e^{-2\kappa t}\Big[\frac{e^{2\kappa u}}{2\kappa}\Big]_0^t = \frac{\sigma^2}{2\kappa}\big(1 - e^{-2\kappa t}\big) \qquad \text{(integrate)}$$

$$r_t \sim \mathcal{N}\Big(r_0 e^{-\kappa t} + \theta(1 - e^{-\kappa t}),\ \frac{\sigma^2}{2\kappa}(1 - e^{-2\kappa t})\Big) \qquad \text{(deterministic integrand} \Rightarrow \text{normal)}$$

where

$\mathbb{E}[r_t]$	weighted average of r_0 and θ; the weight on θ grows from 0 to 1.
$t \to \infty$	stationary distribution $\mathcal{N}(\theta, \frac{\sigma^2}{2\kappa})$ *[Beyond slides]*.
$\kappa \to 0$	variance $\to \sigma^2 t$: back to arithmetic BM *[Beyond slides]*.

[Beyond slides] Weakness: normal, so negative rates possible (CIR $\mathrm{d}r = \kappa(\theta - r)\,\mathrm{d}t + \sigma\sqrt{r}\,\mathrm{d}W$ fixes this). Half-life of a deviation: $\ln 2/\kappa$.

1.11 Implementation *[Beyond slides]* (tested)

```
import numpy as np
from scipy.stats import norm
def bachelier_call(S0, K, sigma, T, r=0.0):
    d = (S0 - K) / (sigma*np.sqrt(T))
    return np.exp(-r*T) * ((S0 - K)*norm.cdf(d) + sigma*np.sqrt(T)*norm.pdf(d))
def bs_call(S0, K, sigma, T, r):
    d1 = (np.log(S0/K) + (r + 0.5*sigma**2)*T) / (sigma*np.sqrt(T))
    d2 = d1 - sigma*np.sqrt(T)
    return S0*norm.cdf(d1) - K*np.exp(-r*T)*norm.cdf(d2)
rng = np.random.default_rng(2); Z = rng.standard_normal(2_000_000)
S0, K, T, r = 100, 105, 1.0, 0.05
ST = S0 + 20*np.sqrt(T)*Z                                  # Bachelier, sigma_N = 20, r = 0
print(round(bachelier_call(S0, K, 20, T), 4), round(np.maximum(ST-K, 0).mean(), 4))  # 5.7269 5.7292
print(round(bachelier_call(100, 100, 20, 1), 4), round(20*np.sqrt(1/(2*np.pi)), 4))  # ATM 7.9788 7.9788
ST = S0*np.exp((r - 0.02)*T + 0.2*np.sqrt(T)*Z)            # Black-Scholes, sigma = 20%
print(round(bs_call(S0, K, 0.2, T, r), 4), round(np.exp(-r*T)*np.maximum(ST-K, 0).mean(), 4))  # 8.0214 8.0257
kappa, theta, sig, r0, t = 1.5, 0.04, 0.01, 0.02, 2.0      # Vasicek
mean = r0*np.exp(-kappa*t) + theta*(1 - np.exp(-kappa*t))  # 0.039
var = sig**2/(2*kappa)*(1 - np.exp(-2*kappa*t))            # 3.33e-5
```

2 Comparison tables

2.1 The three SDEs

	Bachelier [S6–S8]	Black–Scholes [S9–S14]	Vasicek [S15–S16]
SDE	$\mathrm{d}S = \sigma\,\mathrm{d}W$	$\mathrm{d}S = rS\,\mathrm{d}t + \sigma S\,\mathrm{d}W$	$\mathrm{d}r = \kappa(\theta - r)\,\mathrm{d}t + \sigma\,\mathrm{d}W$
Transform	none	$\log S$	$e^{\kappa t} r_t$
Solution	$S_0 + \sigma W_T$	$S_0 e^{(r - \frac{1}{2}\sigma^2)T + \sigma W_T}$	$r_0 e^{-\kappa t} + \theta(1 - e^{-\kappa t}) + \sigma \int e^{\kappa(u-t)}\,\mathrm{d}W$
Distribution	normal	lognormal	normal
Mean	S_0	$S_0 e^{rT}$	$r_0 e^{-\kappa t} + \theta(1 - e^{-\kappa t})$
Variance	$\sigma^2 T$	$S_0^2 e^{2rT}(e^{\sigma^2 T} - 1)$ *[Beyond slides]*	$\frac{\sigma^2}{2\kappa}(1 - e^{-2\kappa t})$
Negative values	possible	impossible	possible
Used for	rates, spreads	equities, FX	short rates

2.2 Bachelier vs Black–Scholes call

	Bachelier	Black–Scholes
x^*	$\frac{K-S_0}{\sigma\sqrt{T}}$	$\frac{\log(K/S_0)-(r-\frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$
Price	$e^{-rT}[(S_0-K)\Phi(d)+\sigma\sqrt{T}\phi(d)]$	$S_0\Phi(d_1)-Ke^{-rT}\Phi(d_2)$
Key step	$\int xe^{-x^2/2}=-e^{-x^2/2}$	completing the square
ATM	$e^{-rT}\sigma\sqrt{T/2\pi}$	$\approx 0.4S_0\sigma\sqrt{T}$ when $r=0$ *[Beyond slides]*
σ units	price	percentage

3 Master formula sheet

- **Itô and martingales [S5]:** $\mathrm{d}g(t,W)=(g_t+\frac{1}{2}g_{xx})\,\mathrm{d}t+g_x\,\mathrm{d}W$; martingale $\iff$ drift $=0$.
- **Bachelier [S6–S8]:** $S_T=S_0+\sigma W_T\sim\mathcal{N}(S_0,\sigma^2T)$; $x^*=\frac{K-S_0}{\sigma\sqrt{T}}$; $V_c=e^{-rT}[(S_0-K)\Phi(d)+\sigma\sqrt{T}\phi(d)]$, $d=\frac{S_0-K}{\sigma\sqrt{T}}$; ATM $e^{-rT}\sigma\sqrt{T/2\pi}$.
- **Black–Scholes [S9–S14]:** $\mathrm{d}\log S=(r-\frac{1}{2}\sigma^2)\,\mathrm{d}t+\sigma\,\mathrm{d}W$; $S_T=S_0e^{(r-\frac{1}{2}\sigma^2)T+\sigma W_T}$; $V_c=S_0\Phi(d_1)-Ke^{-rT}\Phi(d_2)$, $d_{1,2}=\frac{\log(S_0/K)+(r\pm\frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$, $d_2=-x^*$.
- **Vasicek [S15–S16]:** $r_t=r_0e^{-\kappa t}+\theta(1-e^{-\kappa t})+\sigma\int_0^t e^{\kappa(u-t)}\,\mathrm{d}W_u\sim\mathcal{N}\big(r_0e^{-\kappa t}+\theta(1-e^{-\kappa t}),\frac{\sigma^2}{2\kappa}(1-e^{-2\kappa t})\big)$.

4 Glossary

- **Arithmetic Brownian motion:** $\mathrm{d}S=\mu\,\mathrm{d}t+\sigma\,\mathrm{d}W$; normal outcomes [S7]. **Bachelier model:** arithmetic BM for prices (1900) [S6].
- **Black–Scholes model:** geometric BM for prices (1973) [S9]. **Geometric BM:** $\mathrm{d}S=\mu S\,\mathrm{d}t+\sigma S\,\mathrm{d}W$; lognormal [S11].
- **Completing the square:** $x^2-2ax=(x-a)^2-a^2$ to integrate exponentials of quadratics [S13].
- **Exercise boundary x^*:** level of the standard normal above which the option pays [S7, S12].
- **Integrating factor:** multiplier ($e^{\kappa t}$) that removes the state from the drift [S15].
- **Mean reversion / Ornstein–Uhlenbeck / Vasicek:** $\mathrm{d}X=\kappa(\theta-X)\,\mathrm{d}t+\sigma\,\mathrm{d}W$; Vasicek (1977) for the short rate [S15].

5 Self-test

Questions

Q1. ([S5]) Derive the SDEs of the six processes and say which are martingales.

Q2. ([S6]) Under Bachelier, find $\mathbb{P}(S_T<0)$ for $S_0=100$, $\sigma=30$, $T=4$.

Q3. ([S7–S8]) Derive the Bachelier call formula and its ATM value. Price $S_0=K=100$, $\sigma=20$, $T=1$, $r=0$.

Q4. ([S8]) Give the Bachelier put formula.

Q5. ([S11]) Solve $\mathrm{d}S=rS\,\mathrm{d}t+\sigma S\,\mathrm{d}W$.

Q6. ([S12–S14]) Derive $V_c=S_0\Phi(d_1)-Ke^{-rT}\Phi(d_2)$. What is $\Phi(d_2)$?

Q7. $S_0=100$, $K=105$, $T=1$, $r=5\%$, $\sigma=20\%$: price the call and the put.

Q8. ([S10]) Compare Bachelier and Black–Scholes paths.

Q9. ([S15]) Solve the Vasicek SDE.

Q10. ([S16]) $\kappa=1.5$, $\theta=4\%$, $\sigma=1\%$, $r_0=2\%$, $t=2$: mean, standard deviation and $\mathbb{P}(r_t<0)$.

Solutions

S1. See 1.2: martingales are (4) $W^2 - t$ and (6) $e^{\theta W - \frac{1}{2}\theta^2 t}$.

S2. $S_T \sim \mathcal{N}(100, 3600)$: $\Phi(-100/60) = \Phi(-1.667) \approx 4.8\%$.

S3. See 1.3–1.4. ATM: $20\sqrt{1/2\pi} = 7.979$.

S4. $V_p = e^{-rT}[(K - S_0)\Phi(-d) + \sigma\sqrt{T}\phi(d)]$, $d = \frac{S_0 - K}{\sigma\sqrt{T}}$.

S5. Itô on $\log S$: $\mathrm{d}\log S = (r - \frac{1}{2}\sigma^2)\,\mathrm{d}t + \sigma\,\mathrm{d}W$; $S_T = S_0 e^{(r-\frac{1}{2}\sigma^2)T + \sigma W_T}$.

S6. See 1.7–1.8. $\Phi(d_2) = \mathbb{Q}(S_T > K)$.

S7. $d_1 = \frac{\log(100/105) + 0.07}{0.2} = 0.1061$, $d_2 = -0.0939$; $V_c = 100(0.5422) - 105e^{-0.05}(0.4626) = 8.02$. Put $8.02 - 100 + 105e^{-0.05} = 7.90$.

S8. BS moves scale with the price and stay positive; Bachelier moves are level-independent and can go negative.

S9. $r_t = r_0 e^{-\kappa t} + \theta(1 - e^{-\kappa t}) + \sigma \int_0^t e^{\kappa(u-t)}\,\mathrm{d}W_u$.

S10. Mean $0.02e^{-3} + 0.04(1 - e^{-3}) = 0.0390$; variance $\frac{10^{-4}}{3}(1 - e^{-6}) = 3.325 \times 10^{-5}$, sd 0.577%; $\mathbb{P}(r_t < 0) = \Phi(-6.76) \approx 0$.

Appendix A. Slide coverage checklist

Slide	Content	Section
S2–S4	Bru & Yor; process classes; Itô calculus acceptance	1.1
S5	SDEs and martingales: six examples	1.2
S6–S7	Bachelier model; call integral; x^*	1.3
S8	Bachelier call formula; ATM	1.4
S9–S10	Black–Scholes model; sample paths	1.5
S11	Solving GBM via Itô on $\log S$	1.6
S12	BS call integral; x^*	1.7
S13–S14	Completing the square; d_1, d_2	1.8
S15	Vasicek SDE and solution	1.9
S16	Vasicek mean, variance, distribution	1.10

Appendix B. Flag summary (△!)

1. [S5] Zero drift gives a local martingale; true martingale needs integrability (satisfied by all six examples).
2. [S6–S8] Bachelier with zero drift and e^{-rT} discounting is only risk-neutral if $r = 0$ or S is a forward price.
3. [S10] Same σ used for both models; Bachelier σ is in price units ($\approx \sigma_{BS} S_0$).
4. [S6] "Reasonable model for interest rates": yes for allowing negative values, but mean reversion is usually added.
5. [S5, S10] Example working and the comparison answer are not on the slides; the working here is my own.